

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EE)

December 2013

EE-306 ELECTRICAL MACHINES III

Date: 3.12.2013, Tuesday

Time: 1:30 a.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Differentiate between hydro generator and turbo generators [07]

Q - 2 (a) Explain the conditions needed to be satisfied in synchronization of alternator [04]

(b) Why is armature winding kept stationary in synchronous machines? [03]

(c) Define the cording of coil and distribution of coil along with vector diagrams, what is the effect of the same on Emf generated? [07]

OR

Q - 2 (a) Draw equivalent circuit and phasor diagrams of synchronous generator for lagging power factor, leading power factor, and unity power factor [06]

(b) Explain in details the construction of Synchronous Machines. [04]

(c) Derive the Emf equation of a synchronous generator. What are the assumptions made for the same? [04]

Q - 3 (a) Draw and explain synchronous generator capability curve. State the information we get from this curve. [07]

(b) Two three-phase alternators operate in parallel. The rating of one machine is 200 MW and that of the other is 400 MW. The droop characteristics of their governors are 4% and 5% respectively from no load to full load. Assuming that the governors are operating at 50 Hz at no load, how would a load of 600 MW be shared between them? What will be the system frequency at this load? Repeat the problem if both the governors have a droop of 4%. [07]

OR

Q - 3 (a) Discuss the P- δ curve and Q- δ curve of cylindrical rotor SG [05]

(b) A 3 phase Y-connected alternator is rated at 1600 KVA, 13500 V. The armature effective resistance and synchronous reactance are 1.5 ohm and 30 ohm respectively per phase. Calculate the % regulation for a load of 1280 KW at p.f of
i) 0.8 leading. ii) Unity [04]

(c) Explain what do you mean by voltage regulation explain the Emf method of voltage regulation for a Synchronous machine. [05]

SECTION – II

- Q - 4 (a) Select appropriate choices [03]
- 1.) The frequency of voltage generated by an alternator having 8 poles and rotating at 250 rpm is
(A) 60 Hz (B) 50 Hz
(C) 25 Hz (D) $16 \frac{2}{3}$ Hz.
 - 2.) As the speed of an alternator increases
(A) the frequency increases
(B) the frequency decreases
(C) the frequency remains constant but power factor decreases
(D) None of the above.
 - 3.) Alternators used in aircraft systems usually have frequency of
(A) 25 Hz
(B) 50 Hz
(C) 100 Hz
(D) 400 Hz.

- (b) a.) Discuss the effect of increase in Mechanical input of Generator. [04]
b.) Discuss the effect of increase in excitation of Generator.

- Q - 5 (a) Explain the effect of varying the load of the synchronous motor and keeping the excitation constant, along with the necessary vector diagram. And explain the 'O' curve along with the conclusion derived from the same. [06]
- (b) Draw and Explain the 'V' curve and inverted 'V' curve, of the synchronous machine. [05]
- (c) Draw and explain the power flow diagram of synchronous motor. [03]

OR

- Q - 5 (a) Explain the effect of varying the field excitation while keep the load on synchronous motor constant. Also draw the necessary vector diagram. [06]
- (b) Explain the phenomenon of hunting or phase swinging in synchronous motors. [04]
- (c) Write a note on cooling systems of synchronous machines. [04]
- Q - 6 (a) Explain the concept, construction, working and advantage of switched reluctance motor. [07]
- (b) Explain the concept, construction, working and advantages of PMSM. [07]

OR

- Q - 6 (a) Write a short note on Induction Voltage Regulator. [05]
- (b) Derive the dynamic (swing) equation of synchronous machine. [05]
- (c) Write a short note on PMSM [04]
